Answer **all** the questions in this Section in the spaces provided.

1 A spring is attached at one end to a fixed point and hangs vertically with a cube attached to the other end. The cube is initially supported at a height *h* above the water surface such that the spring is at its natural length, as shown in Fig. 1.1.

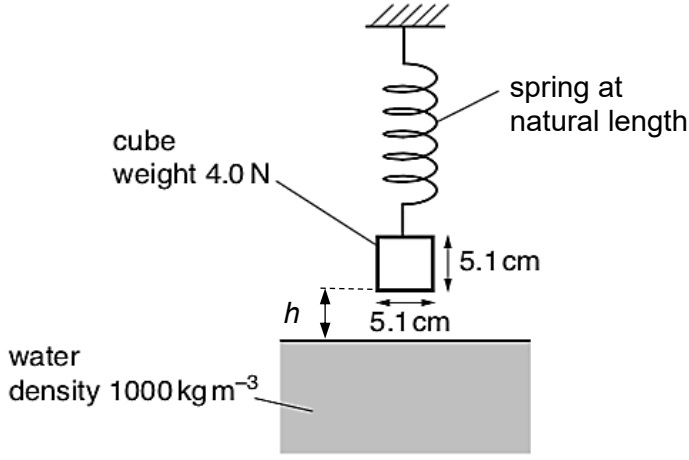

Fig. 1.1

The cube has a weight of 4.0 N and sides of length 5.1 cm. The cube is lowered gently into water. The cube reaches equilibrium with its base at a depth of 7.0 cm below the water surface, as shown in Fig. 1.2. The density of the water is 1000 kg m^{-3}.

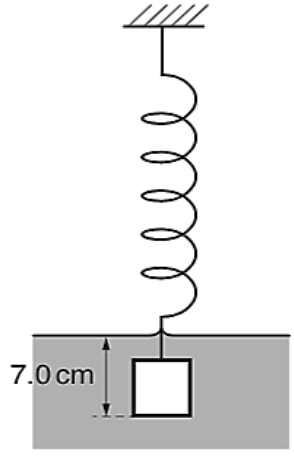

Fig. 1.2

(a) Calculate the difference in the pressure exerted by the water on the bottom face and on the top face of the cube.

difference in pressure = ……. Pa [2]

(b) Using your answer in **(a)**, show that the upthrust acting on the cube is 1.3 N. [1]

(c) Determine the force exerted on the spring by the cube when it is in equilibrium in the water.

force = ... N [2]

(d) The spring obeys Hooke's law and has a spring constant of 30 N m^{-1}.

Determine the height *h*.

h = ... m [3]

(e) The cube in the water is suddenly detached from the spring.

(i) Determine the initial acceleration of the cube.

initial acceleration = ... m s^{-2} [2]

(ii) Describe and explain the variation of the acceleration of the cube as it sinks in the water.

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..... [2]